

G04 / ASSESSMENT

G04 GATE A

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Read this first

Pipeline Engineering boundary after M10

This is a fresh stage-boundary assessment. Use learner books/reviews closed. Save code/config/results/errors before repair. No forced countdown timer.

Dataset: `practice/assessment/practical/water_meter_batches.csv`

1. Architecture and boundaries

Design source -> landing/staging -> serving plus Docker/Airflow control boundaries for `water_meter_batches.csv`. State one-row grain and what remains immutable.

2. Container design

Draft image/Compose structure for a small validator. Use service-name communication, one deliberate port only if host access is needed, and persistent output without committing secrets.

3. Partitioned orchestration

Create daily partitions from `event_date`. Explain data-interval-derived paths, dependencies, retries and task communication.

4. Reconciliation

Compute daily row counts/liter totals and an all-source control. Explain how publish is blocked on invalid/duplicate meter events.

5. Failure/recovery

Describe evidence for one transient retry and one deterministic data failure. Then explain safe reprocessing of Sep2 only.

6. Operational defense

Give a 5-minute verbal handoff: setup, configuration/secrets, logs, sensor/wait, backfill, validation and remaining runtime limits.

Gate decision

PASS only when every critical competency has defensible evidence and no unresolved data-loss, duplicate-side-effect, secret-leak, partition-time or unsafe-retry failure remains. Repair a weak competency, then use the different fresh Gate B after review and repair. Reading Review A is not a Gate B pass.

Path rule: use `09_GATES/G04` as the working folder. The named practice/assessment file is included beneath this gate folder.

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